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Head Distillation for Dental Multiple-Choice Question Answer

Decision-Space Distillation for Deployable Medical QA

DeepSeek-V3 Teacher | Qwen2.5-14B Student | CMExam Full-Test 991 Questions

Key Result: 89.10% student accuracy > 87.18% teacher accuracy

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Presentation Outline

1. Motivation and problem setting
2. Research questions and Choice-Head method
3. Why decision-space distillation matters
4. Experimental setup and benchmark scale
5. Main results and student-over-teacher finding
6. Contributions, discussion, and conclusion

01 Motivation and Problem Setting

Goal: preserve exam performance while making dental MCQ systems smaller, cheaper, and easier to deploy.

Current Tension

- Medical LLMs are accurate on exam benchmarks
- But inference cost and hardware demand remain high
- Deployment value matters beyond raw benchmark scores

Task Structure

- Dental QA is a five-option MCQ task
- Decision space is small and well-structured
- Open-ended generation is not required here

Mismatch

- Full-vocabulary distillation is redundant
- API teachers rarely expose internal logits
- Task-aligned supervision is needed

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Research Questions

RQ1

What to Distill?

Whole vocabulary or only A/B/C/D/E?

RQ2

Can Students Exceed Teachers?

Under a task-aligned objective

RQ3

Can It Stay Black-Box Friendly?

Compatible with API teachers

Central research question

For a five-option medical exam task, the important transfer target may be the decision structure rather than the full language-model distribution.

If the supervision target is redesigned around the downstream task, smaller students may become both deployable and unexpectedly strong.

03 Core Method: Choice-Head Distillation

Teacher

DeepSeek-V3

Can be local or API-based

Provides option-level preference

A/B/C/D/E

Choice Head

Five-option distribution

Decision-space supervision

No full vocabulary needed

KL + CE

Student

Qwen2.5-7B / 14B

Learns the final choice boundary

More deployable

Stage 1 objective

L = alpha * KL(p_T || p_S) + (1 - alpha) * CE

The student learns relative preferences across the five answer options instead of the entire token vocabulary.

Main setting: one Stage 1 epoch with alpha = 0.35 for the best 14B run.

Design intuition

Keep the part of teacher uncertainty that directly affects the final option.
Drop supervision dimensions that do not matter for a five-choice decision.

This is why the method stays lightweight while remaining competitive.

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Why Choice-Head Matters

Alignment

Task-Specific

Supervision follows the five-option structure

Efficiency

Lower Cost

Less redundant training and memory usage

Teacher Access

Black-Box Friendly

Works with API-only teachers

Transfer Signal

Useful Uncertainty

Preserves option-level confusion structure

The method changes distillation from a generic language-model compression problem into a task-specific decision transfer problem.

Better aligned supervision often matters more than a more complicated pipeline.

For structured MCQs, the decision-space target is a cleaner and more deployable abstraction.

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Experimental Setup

Dataset

6,591 Questions

CMExam-based single-choice resplit

Splits

4608 / 991 / 991

Train / Val / Test

Students

Qwen2.5 7B / 14B

LoRA fine-tuning

Teacher

DeepSeek-V3

87.18% on the 991-question test

Evaluation scope

Main benchmark: 991-question full-data test set
Dental subset retained for specialty-focused validation
Large-scale resplit makes the main conclusion more stable

Training setup

LoRA rank = 16, LoRA alpha = 32
Best 14B setting uses one Stage 1 epoch
Main learning rate: 1e-4
Stage 2 is tested but not required for the best 14B result

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Main Results

Teacher

87.18%

DeepSeek-V3 on full-data test

14B Zero-Shot

83.55%

Base student without distillation

14B Distilled Mean

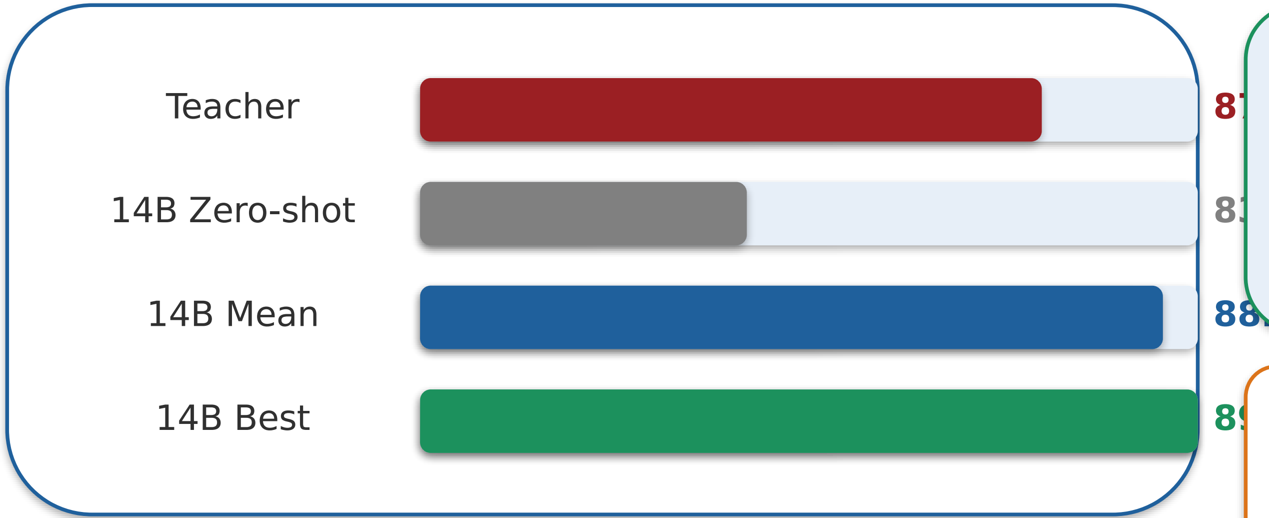
88.67%

Three-seed average

14B Distilled Best

89.10%

Best single run



Takeaway

The distilled student gains 5.12 percentage points over the 14B zero-shot baseline on average.
The best 14B student exceeds the teacher on the same 991-question benchmark.
The 7B student also improves strongly, so the method is not tied to one model

How to read Table 1

Full Test = 991-question main benchmark; Dental Test = 125-question subset. Zero-shot = no distillation; Mean = average runs; Best = strongest single run.

+1.92 points vs teacher

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Two Main Contributions

Contribution 1: Method

Introduces a task-aligned decision-space distillation framework for five-option medical MCQs.
Replaces generic vocabulary-space supervision with a cleaner option-level target.
Keeps the framework practical for both local and API teachers.

Contribution 2: Empirical Finding

Shows that a smaller student can outperform a stronger teacher under the task-aligned formulation.
Validates the result on a 991-question test set rather than only a tiny benchmark slice.
Frames the student-over-teacher effect as a consequence of better task alignment, not a distillation failure.

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Discussion and Interpretation

Stage 2 is Not Always Better

- Helpful for some smaller students
- Can mildly hurt a strong 14B student
- Extra GT fine-tuning may erase soft-label structure

Why the Student Can Win

- Teacher soft labels provide structure
- Student is still optimized for the benchmark
- Task-aligned supervision sharpens the decision boundary

Broader Message

- More complexity is not the point
- The supervision target must fit the task
- Decision-space transfer can be a practical design rule

For structured medical multiple-choice tasks, the critical design choice is not how much of the teacher to copy, but which part of the teacher signal actually matters.

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Conclusion

Choice-Head distillation makes smaller dental MCQ models both practical and highly competitive.

e supervision is an effective alternative to full-vocabulary distillation for structured MCQs.

student reaches 89.10% and surpasses the 87.18% DeepSeek-V3 teacher on the 991-question test se

k stays lightweight, deployment-oriented, and compatible with black-box teachers.

Thank you. Questions are welcome.

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Q&A Backup

Why distill only five options?

The downstream task is a five-option decision problem.

Full-vocabulary supervision introduces unnecessary dimensions.

Option-level transfer is more task-aligned and more efficient.

Why can the student outperform the teacher?

The student learns teacher soft-label structure.

It is still optimized directly for the benchmark objective.

Task-aligned supervision can yield a stronger decision boundary.

Can this extend to open-ended QA?

Not directly.

The current method is built for structured MCQs.

Open-ended QA would require a different supervision target.